MILITARY SPECIFICATION

BAROGRAPH ML-563( )/UM

This specification has been approved by the Department of Defense and is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of equipment designated as Barograph ML-563( )/UM which is a portable precision instrument that is used to measure and register changes in atmospheric pressure over a range of 2-1/2 inches of mercury.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Box, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal and Electronic Test Equipment

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Description. - Barograph ML-563( )/UM measures and records changes in atmospheric pressure by means of an aneroid bellows sensing element which activates a pen over Chart ML-236 secured to the drum of Clock ML-145. Each graduation on the chart represents 0.02 inches of mercury and the chart covers a range of 2.5 inches of mercury. The barograph operates over a range of 2-1/2 inches of mercury adjustable between 8 and 32 inches.

3.2 Preproduction samples. - The contractor shall furnish two preproduction samples for approval, if required by the invitation for bids and contract. (See 4.3.)

3.3 Procurement model. - A procurement model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bids, physical construction of the equipment shall conform to the model and the equipment shall incorporate all features of the model. (NOTE: In case of conflict between specified performance characteristics for the equipment and the performance of the model, the specified performance characteristics govern.)

3.4 Parts, materials, and processes; general. - Parts, materials, and processes shall be such as to enable the equipment to meet its performance and other requirements, as specified herein. Ratings and derated values of parts and materials shall not be exceeded when the equipment is subjected to specified service conditions. (See 3.10.) Derating methods used shall be as specified by the applicable individual documents covering the parts and materials.

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3.5 Finish, protective. - Equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch up after mounting. (See 4.4.)

3.5.1 Paint finish. - The paint finish shall be as per model.

3.6 Screws, nuts, and other threaded parts. - Screw threads shall be in accordance with National Bureau of Standards Handbook H28 covering Screw-Thread Standards for Federal Services, and shall be Class 2 fit unless otherwise specified. Wherever practicable, screw threads shall be commercial standard sizes. Threads in soft material shall be in the American National Coarse-Thread Series; threads in other material shall be in either the American National Coarse-Thread or Fine-Thread Series, as suitable for the particular application.

3.7 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.13.) Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.8 Marking. -

3.8.1 General. - Marking shall conform to Specification MIL-M-13231. (See 4.4.) Front panel marking shall be group I as described in that specification.

3.9 Serial numbers. - Serial numbers are required for the Barograph ML-563( )/UM.

3.10 Service conditions. - The barograph shall meet the requirements of this specification after subjection to each of the following service conditions:

(a) Operation. - The barograph shall be capable of operating continuously at any point in the temperature or humidity operating ranges specified under (b) or (c), below, or any combination thereof, for a period of 30 days.

(b) Temperature. - The barograph shall be capable of operating at any temperature in the range of +32°F. to +132°F., and shall be capable of withstanding continuous exposure to any temperature in the range of -80°F. to +160°F., as determined by the test of 4.8.

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(c) Moisture-resistance. - The barograph shall be capable of operating at humidities up to 95 percent as determined by the test of 4.9.

(d) Elevation. - The barograph shall be capable of operating at any elevation up to 6,000 feet above sea level, and shall withstand storage at any elevation up to 10,000 feet above sea level, as determined by the tests of 4.11 and 4.11.1.

3.11 Mechanical characteristics. -

3.11.1 Clock ML-145. - Clock ML-145 shall run for a period of one week at any temperature between +32°F. and +132°F. A means shall be provided for adjustment of the clock speed so that it may be increased or reduced. The peripheral motion of the cylinder shall be 2.555 inches per day ±.005 inches. The rate of peripheral motion shall be obtained with the clock adjustment within the middle third of its range of adjustment. This rate of peripheral motion has been specified so that with the use of Barograph Chart ML-236, a continuous record of pressure during a period of four days may be obtained without interference with the chart clip.

3.11.1.1 Chart clip. - The clip, which is used to hold the chart on the clock, shall be in accordance with the model.

3.11.1.2 Exception to the model. - The details of the mechanisms of Clock ML-145 may differ from the model provided all requirements of the specification applicable thereto are met.

3.11.2 Pressure element. - The element shall be free from metallurgical defects, properly heat-treated and aged, well made and symmetrical. This element together with the linkage mechanism shall function properly at any pressure between 23.5 and 31.0 inches of mercury, and when referenced to the standard line spacing of Barograph Chart ML-236, at room temperature and through a pressure range of 2-1/2 inches of mercury (both decreasing and increasing) the barograph shall record pressure accurately to within ±.02 inches of mercury.

3.11.2.1 Range of adjustment. - The barograph shall be capable of adjustment so that it may record pressure between 8 and 32 inches of mercury in steps of 2.5 inches.

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3.11.2.2 Temperature compensation. - The instrument shall not indicate a change in pressure greater than .035 inches when subjected to temperatures between +32°F. and +132°F. while held at a constant pressure of 30 ±.50 inches of mercury, without recourse to change in linkage magnification.

3.11.2.3 Pen arm. - The pen arm shall be in accordance with the model. The pressure of the pen shall be adjustable so that no evidence of sticking is apparent.

3.11.2.4 Pen. - The pen shall be in accordance with the model.

3.11.3 Gears. - Gears shall be properly aligned and meshed and shall operate without interference, tightness, looseness, or other irregularities.

3.12 Technical literature, tools, and running spare parts. - Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2.)

3.13 Workmanship. - The equipment will be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.4 Parts, materials and processes
3.5 Finish
3.8 Marking

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. - Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3.)

(b) Inspection covered by subsidiary documents. (See 4.4.)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5.)

(2) Quality conformance inspection of preparation for delivery. (See 4.14.)

4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspection specified for Group A and Group B. (See Tables II and III.)

Table I Preproduction inspection

Inspection	Rqt Par	Test Par
Pressure	3.10	4.11
Time	3.11.1	4.10
Temperature	3.10	4.8
Moisture-resistance	3.10	4.9

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.5
Marking	3.8

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4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4, 4.5.1 through 4.5.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.4.) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. - This inspection, including sampling, shall conform to Table II and the inspection procedures of Standard MIL-STD-105, using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the Government, except the operation inspection (4.7) shall be last.

Table II Group A inspection

Inspection	Req Para.	Test Para.	AQL Major	Minor
Operational		4.7	1.5%	4.5%
Visual-Mechanical		4.6	1.5%	4.5%

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table III and to the procedures for small-sample inspection of Standard MIL-STD-105, using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S-4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

Table III Group B inspection

Inspection	Req. Para.	Insp. Para.
Accuracy of Clock ML-145	3.11.1	4.10
Pressure Element	3.11.2	4.11
Temperature Compensation	3.11.2.2	4.12
Interchangeability	3.7	4.13

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4.6 Visual and mechanical inspection. - Parts and equipment shall be inspected for workmanship, mechanical fit, loose nuts and screws, and miscellaneous defects. Controls and fastening devices shall be inspected for mechanical operation. Soldered connections, welds, finishes, etc. shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required. Examination also may be made for other visual or mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government as major or minor in accordance with the definitions of Standard MIL-STD-105.

4.7 Operational examination. - The barograph shall be operated at room temperature and pressure for a period of five minutes to ensure satisfactory operation of the equipment.

4.8 Temperature tests for equipment. -

4.8.1 Equipment operating and non-operating. - The barograph shall be subjected to the applicable temperature cycle shown on Standard MIL-STD-169, to determine compliance with 3.10(b). The barograph shall meet full specification requirements when measurements are made at the first, fifth, and tenth steps of the referenced standard.

4.9 Moisture-resistance test for equipment. - The barograph shall be tested as follows:

a. Dry at 130° ± 5°F. for 24 hours.

b. Conditions at 77° ± 5°F. and 40 to 50 percent relative humidity for 24 hours.

c. Subject to continuous cycling for two 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170.

d. After cycling has been completed, condition the equipment for 24 hours at 77°F. ± 5°F. and 40 to 60 percent relative humidity. No repair or replacement of parts shall be made, and the barograph shall be capable of meeting full specification requirements.

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4.10 Accuracy of Clock ML-145. - At room temperature (60°F. to 80°F.) the rate of peripheral motion of clock cylinder as specified in 3.11.1 shall be checked to assure that the required accuracy is obtained. If this measurement is not made by means of a "Watchmaster" as made by American Time Products, Inc., or equal, the test shall be conducted over a period of 72 hours.

4.11 Pressure element test. - Barograph ML-563/UM shall be tested for compliance with the requirements of paragraph 3.10(d). Barograph ML-563/UM shall be placed in a pressure chamber at room barometric pressure. The pressure chamber shall then be sealed and the chart value compared with the true pressure indication of the pressure inspection standard. The pressure shall then be raised or lowered, as may be required, in steps of 0.5 inches ± 0.05 inches, with a pause of five minutes or more at each step to allow a decrease in the hysteresis effect on the element, through the full operating pressure range of 2.5 inches. The change in chart readings as indicated by the barograph pen shall equal the change in pressure readings as indicated by the pressure inspection standard to within ±0.02 inches of mercury throughout the entire 2.5 inch range.

4.11.1 Pressure non-operating test. - Barograph ML-563/UM shall be subjected to a pressure of 700 millibars, for a period of 8 hours. At the end of this period, the pressure shall be returned to ambient, and the instrument shall be allowed to rest for 15 hours. Upon completion of this test, the barograph shall meet the requirements of 3.11.2.

4.12 Temperature compensation. - Barograph ML-563( )/UM shall be placed in a pressure chamber and tested for compliance with 3.11.2.2. Readings shall be made after equipment has reached thermal stability through the temperature range of +32°F. through +132°F.

4.13 Inspection for interchangeability. - The equipment shall be inspected for conformance to 3.7.

4.14 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Level A. - Packaging shall be developed for the ML-563( )/UM in accordance with the methods listed below per MIL-P-116, employing kinds and types of materials which when applied will afford adequate protection against corrosion, deterioration and damage during world-wide shipment, handling and open storage:

Technical literature - 1C-6, Barograph-11b, Clock Cylinder-11c and spare pen-111.

5.1.2 Level C. - ML-563( )/UM, shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. -

5.2.1 Level A. - ML-563( )/UM, shall be packed within nailed wood, wood cleated plywood or fiberboard boxes conforming to the requirements of PPP-B-621, PPP-B-601 or PPP-B-636. Shipping containers shall be closed and strapped as specified in the applicable box specification. The gross weight of nailed wood and wood cleated plywood containers shall not exceed 200 pounds, fiberboard containers shall not exceed 65 pounds.

5.2.2 Level B. - ML-563( )/UM, shall be packed as specified in 5.2.1, except containers shall be domestic class.

5.2.3 Level C. - ML-563( )/UM, shall be packed within shipping containers in a manner that will afford adequate protection against damage to the packages and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking. - Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. - Barograph ML-563( )/UM is used to obtain a continuous recording of Atmospheric Pressure (see 1.1).

6.2 Ordering Data. - Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment (Level A, Level B, or Level C).

(d) Samples required (see 3.2).

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical Literature required (see 3.12).

(h) Quantity of tools and running spares required (see 3.12).

6.3 Nomenclature. - The parenthesis in the nomenclature will be deleted and replaced by a letter identifying the particular design; for example: ML-563W/UM. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.4 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

MIL-B-55298

CUSTODIAN:
ARMY-EL
AF-71

REVIEW ACTIVITIES:
AF-71

PREPARING ACTIVITY
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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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